JSP Programs Database Connectivity with MySQL

To connect Java application with the MySQL database, we need to follow 5 following steps.

In this example we are using MySql as the database. So we need to know following informations for the mysql database:

1. **Driver class:** The driver class for the mysql database is **com.mysql.jdbc.Driver**.
2. **Connection URL:** The connection URL for the mysql database is **jdbc:mysql://localhost:3306/sonoo** where jdbc is the API, mysql is the database, localhost is the server name on which mysql is running, we may also use IP address, 3306 is the port number and sonoo is the database name. We may use any database, in such case, we need to replace the ***cse*** with our database name.
3. **Username:** The default username for the mysql database is **root**.
4. **Password:** It is the password given by the user at the time of installing the mysql database. In this example, we are going to use root as the password.

Let's first create a table in the mysql database, but before creating table, we need to create database first.

1. create database cse;
2. use cse;
3. create table studentname (firstname varchar(80), lastname varchar(80));

Example to Connect Java Application with mysql database

Files creation

Step 1: Open the path C:\xampp\tomcat\webapps\ROOT

Step 2: Create One folder Name Ex; csea

Step 3: open C:\xampp\tomcat\webapps\ROOT\csea

Step 4: Here create our files example

form.jsp and reg.jsp

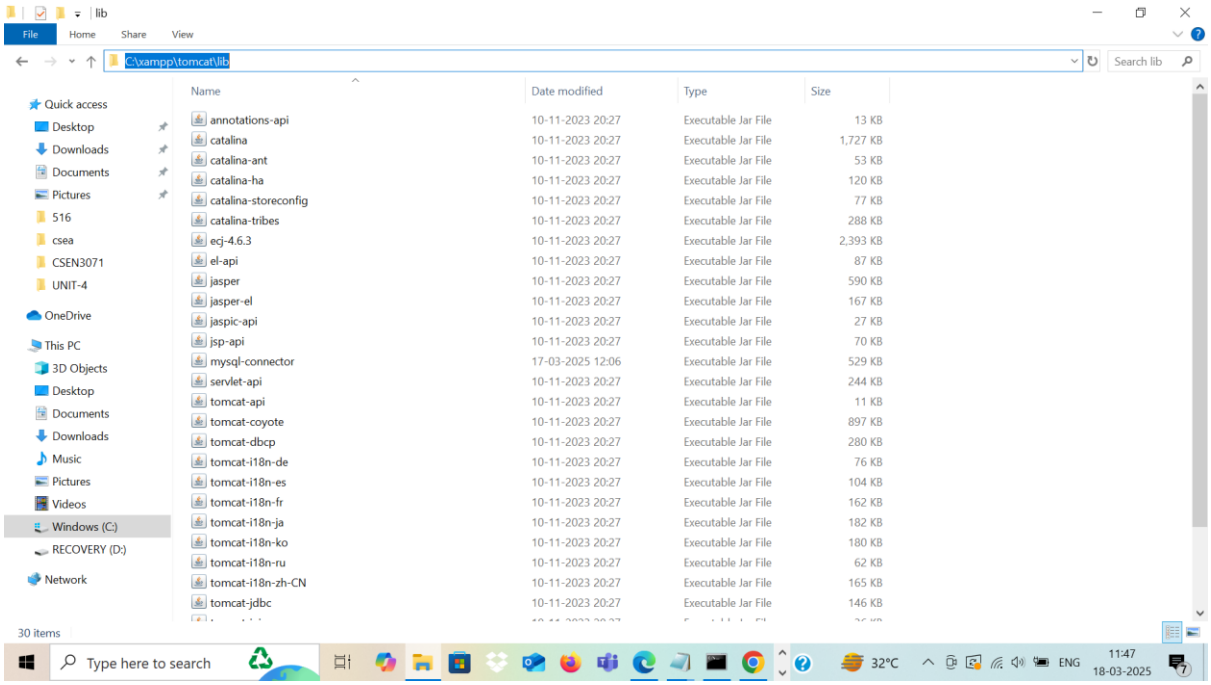
Database Setup Connection with Tomcat Server

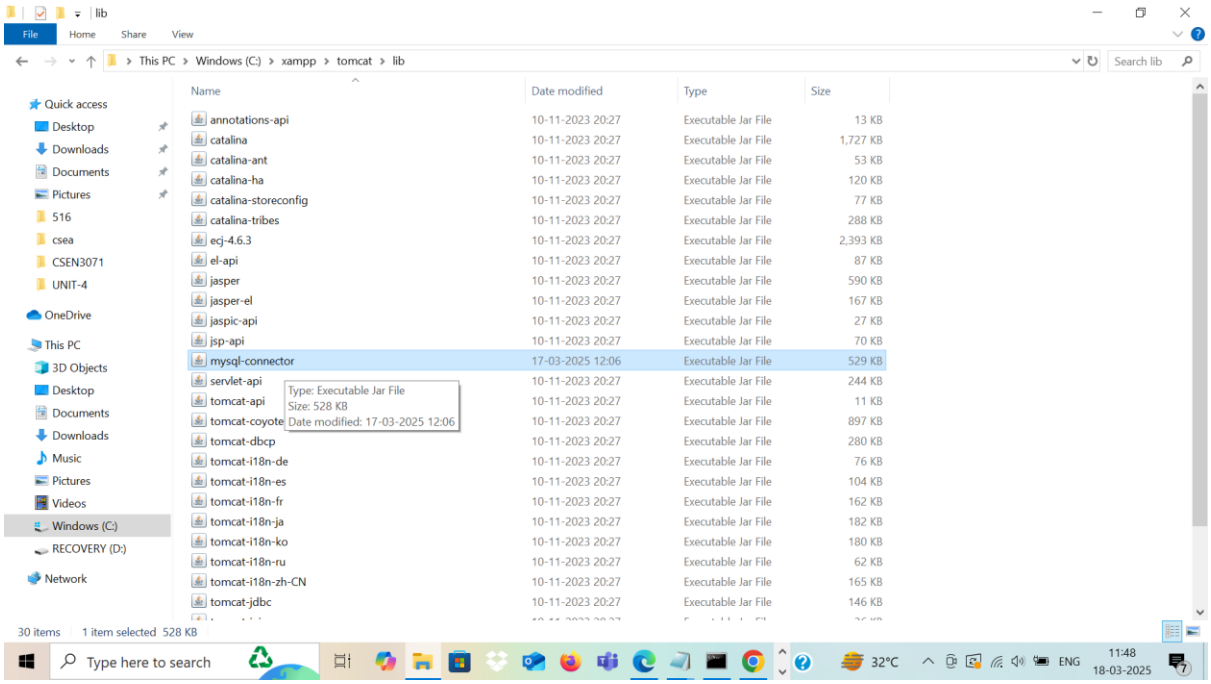
To connect JSP (java) application with the mysql database, **mysqlconnector.jar** file is required to be loaded.

download the jar file mysql-connector.jar

C:\xampp\tomcat\lib

Copy mysql-connector.jar file to above *lib* folder





Source code:

form.jsp

```
<html>

  <body>

          <form action = "reg.jsp" method = "GET">

       first name: <input type = "text" name = "fname">

       <br/>

       lastname: <input type = "text" name = "lname" >                    <br/>

       <input type = "Submit" value = "insert" />

     </form>

  </body>

</html>
```

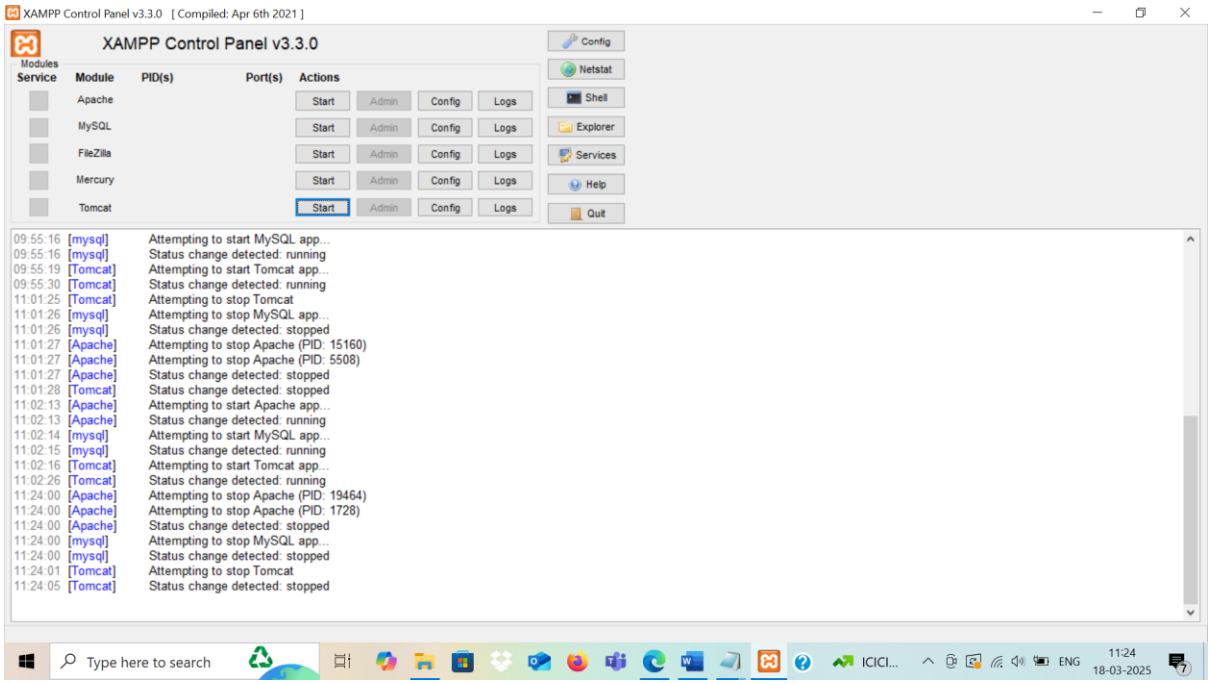
reg.jsp

```
<%@ page import ="java.sql.*" %>
```

```
<%

    String first_name = request.getParameter("fname");

    String last_name = request.getParameter("lname");

    Class.forName("com.mysql.jdbc.Driver");

  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/cse", "root", "");

    Statement st = con.createStatement();

    //ResultSet rs;

   st.executeUpdate("insert into studentname(firstname,lastname)values('" + first_name + "','" + last_name + "')");

     out.print("Registration Successfull!"+"<a href='loginform.jsp'>Go to insert again</a>");



%>
```

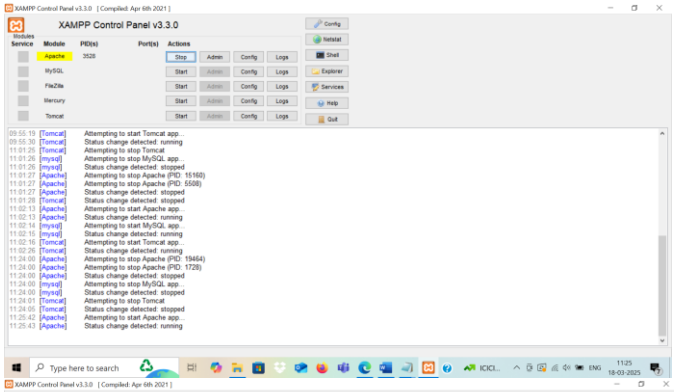
Execution Procedure

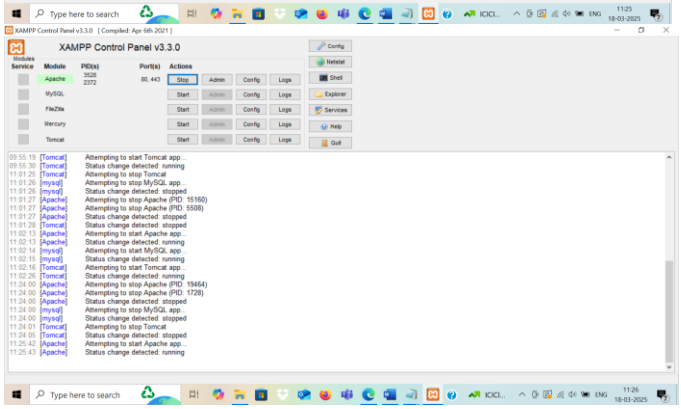
Step 1: Open xampp control panel



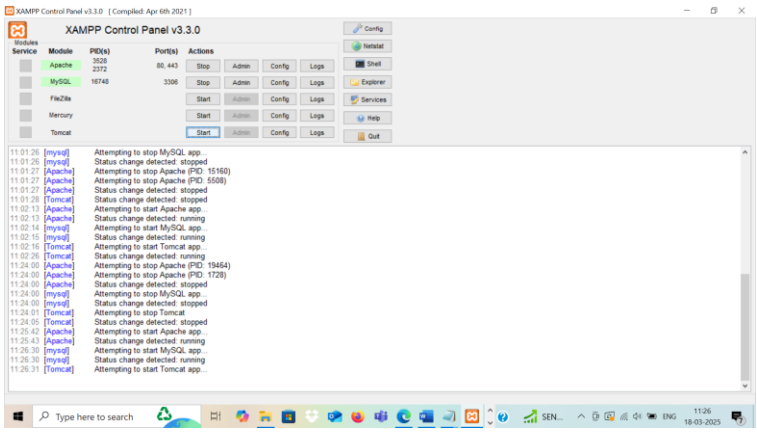
Run

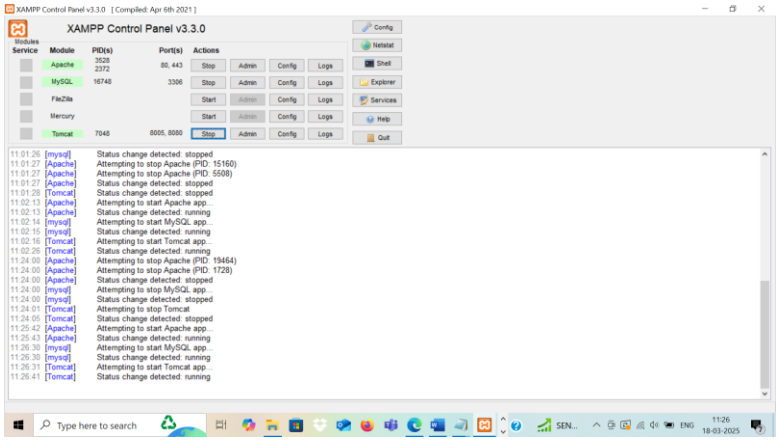
Apache Press Start button



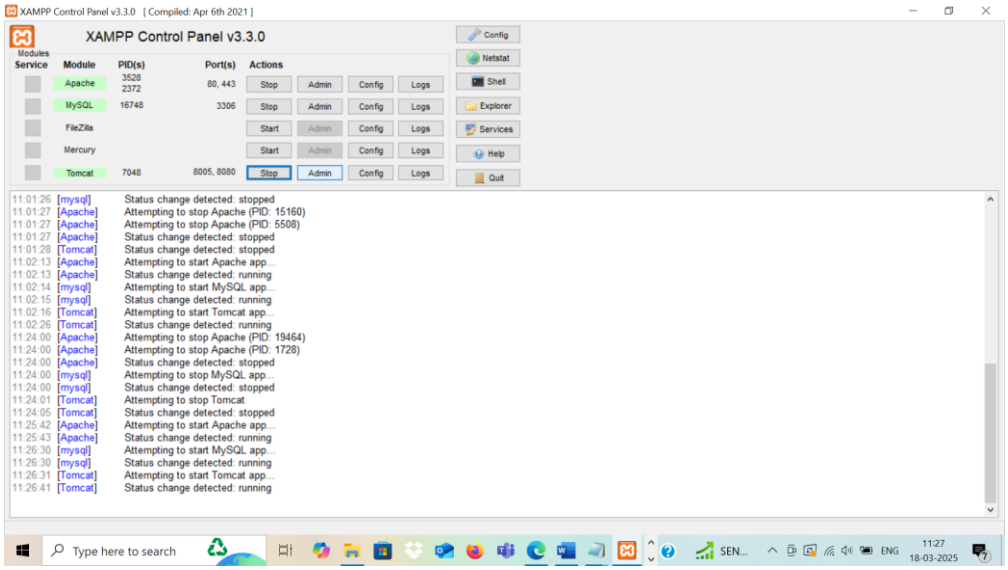


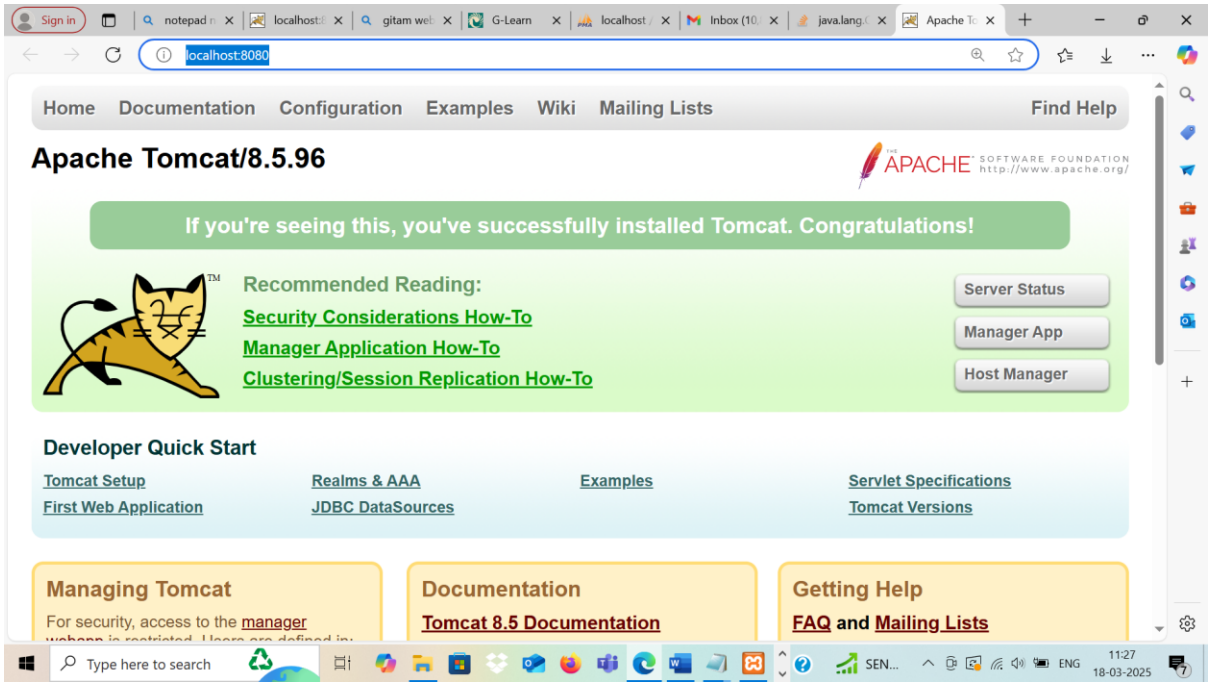
Similary MySql and Tomcat



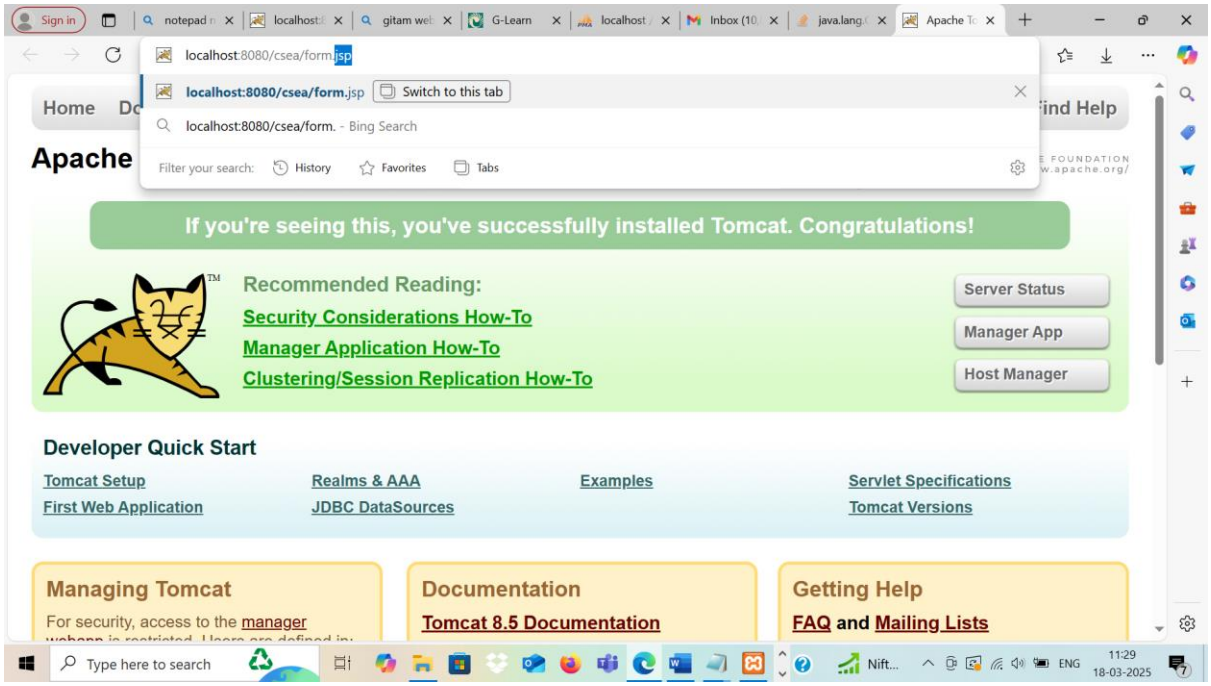


Now Press Admin button for Tomcat

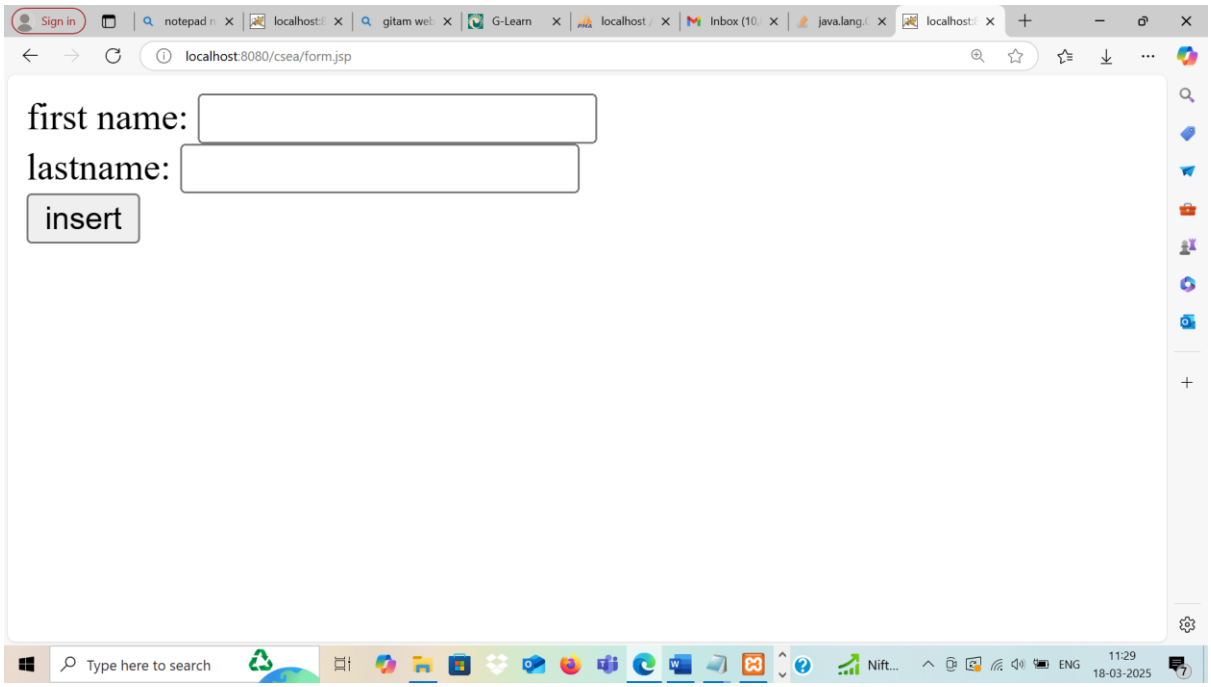




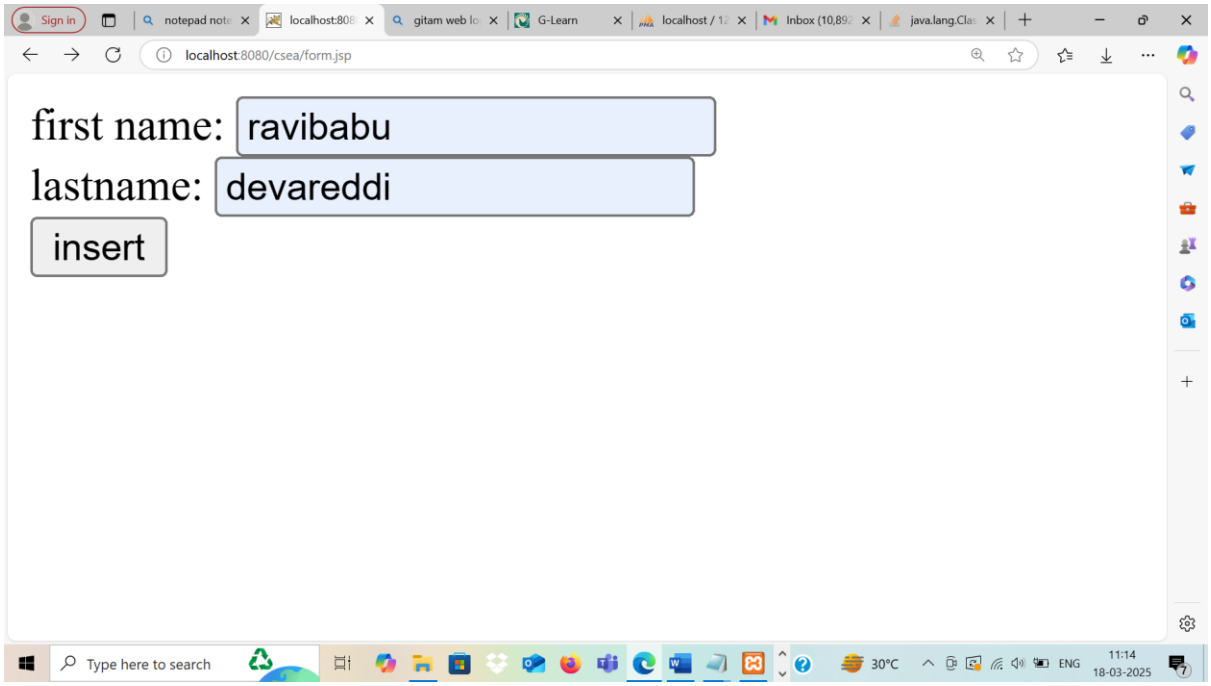
Type the path in the above browser

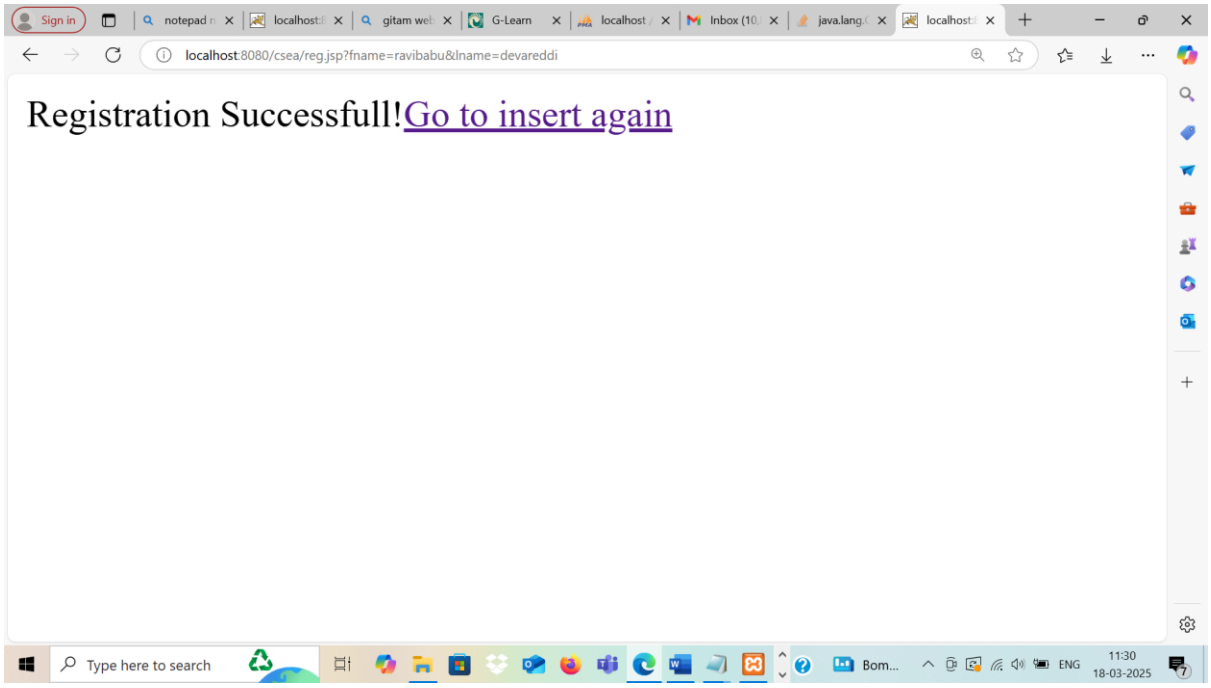


localhost:8080/csea/form.jsp



Enter data to the files and submit button





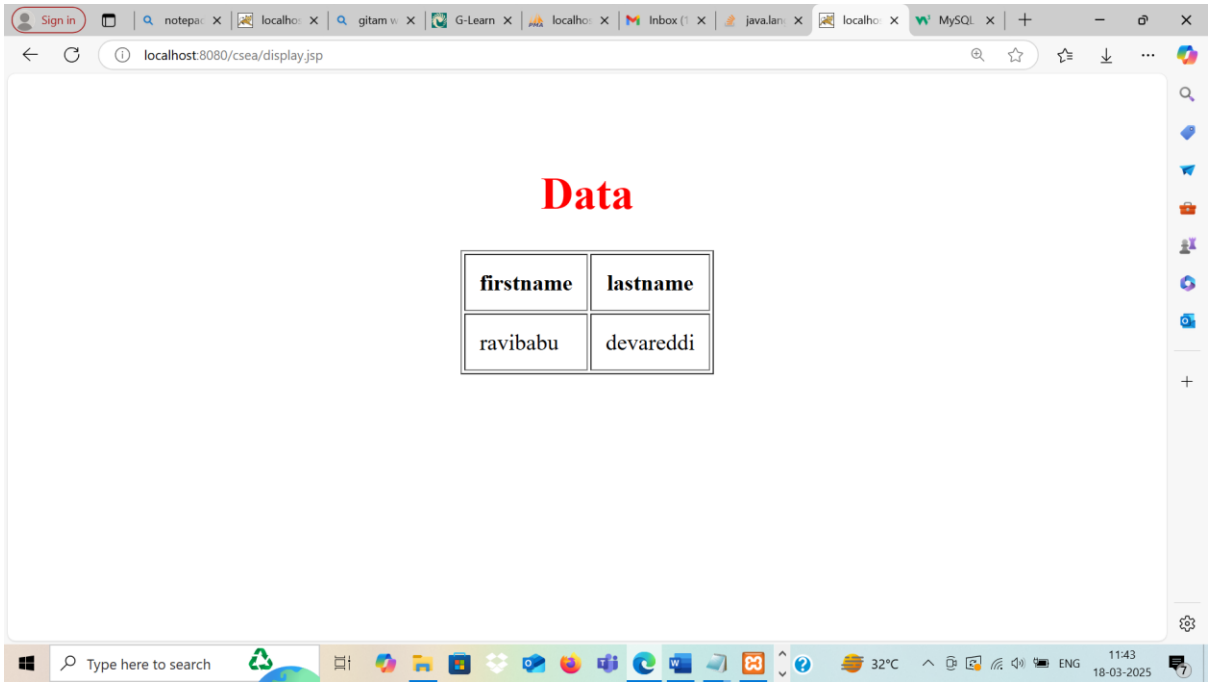
JSP program to retrieve data from MYSQL database table

display.jsp

```
<%@page import="java.sql.*" %>
<%@page import="java.io.*" %>
<html>
<body>
</br>
```

```
</br>
<h1 align='center' style="color:red">Data</h1>



<form action='display.jsp'>
<table align='center' cellpadding='10' border='1'>


<%
try
{

    Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/cse", "root", "");

Statement st=con.createStatement();
ResultSet rs=st.executeQuery("select firstname,lastname from studentname");
ResultSetMetaData rsmd = rs.getMetaData();

for(int i=1 ; i<=rsmd.getColumnCount() ; i++)
{
out.print("<th>"+rsmd.getColumnName(i)+"</th>");
}

while(rs.next())
{

out.print("<tr>");

for(int i=1 ; i<=rsmd.getColumnCount() ; i++)
{
out.print("<td>"+rs.getString(i)+"</td>");
}
out.print("</tr>");
}
}
catch(Exception e)
{
out.print(e);
}
%>
</table>

</form>
</body>
</html>
```

localhost:8080/csea/display.jsp


localhost:8080/csea/display.jsp
Data
firstname
lastname
ravibabu
devareddi
Type here to search
32°C
ENG
11:43
18-03-2025